

# INTRODUCTION TO ANESTHESIA II

## The pre operative assessment

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# Unit one

## The pre operative evaluation

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- Pre anesthetic evaluation/ The preoperative evaluation is evaluation of the patients medical physical and mental status before surgery.
- The preoperative evaluation serves as a screening tool to anticipate and avoid airway difficulties or problems with anesthetic drugs.
- In addition to the history and physical examination, previous anesthesia records should be reviewed and contraindications to specific drugs, such as succinylcholine, nitrous oxide, or volatile agents, should be sought.

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- Pre-operative visits are conducted in a timely manner to provide standardized, essential, evidence based pre-operative information to anesthesia and surgical care teams.
  - Preparing a patient for anesthesia requires an understanding of the patient's pre-operative status, understanding the nature of the surgery, the anesthetic techniques required for surgery as well as the risks that a particular patient may face.
  - Pre-operative anesthetic assessment services can decrease cancellation on the day of surgery, improve the patient's experience of their hospital admission, reduce complication rates and mortality and relieves anxiety.
  - A good anesthetic begins with a good plan.

# Aim of pre operative assessment

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- To assess the patient fitness for anesthesia.
- To reduce the risks associated with surgery and anesthesia.
- To increase the quality of care and decrease the cost.
- To restore the patient to the desired level of function.
- To obtain the patients informed consent for the anesthetic procedure

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- Documentation of the conditions for which surgery is needed.
  - Assessment of the patient's overall health status.
  - Uncovering of hidden conditions that could cause problems both during and after surgery.
  - Peri-operative risk determination.
  - Development of an appropriate peri-operative care plan.

## Preoperative visit

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- Is used to provide an explanation and reassurance for the patient.
- It determine any potential interaction between concurrent disease and anesthesia.
- Reduction of costs, shortening of hospital stay reduction of cancellations and increase of patient satisfaction.
- All patients would be seen by their anesthetist a head of the planed surgery

# Methods of assessment

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- History and physical examination are the most important assessors of disease and risk.
- Necessary pre-operative laboratory investigations are supplemental to this.
- 56% of correct diagnoses were made with the history alone, which increased to 73% with the addition of physical examination.
- Diagnostic tests such as chest radiographs and electrocardiograms(ECG) helped with only 3% of diagnoses.

# Steps of pre operative visit

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1.Problem identification

2.Risk assessment

3.Pre operative preparation

4.Plan of anesthetic technique

# Problem identification

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- History
- Physical examination
- laboratory studies
- Anesthetic note
- ASA classification
- NPO guidelines
- Pre medication

# History in general

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- Identifying Data; such as name, age, gender, occupation, marital status.  
....
- Source of the history; usually the patient, but can be family member etc
- Reliability; Varies according to the patient's memory, trust, and mood.

# History

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- Chief Complaint; The one or more symptoms or concerns causing the patient to seek care.
- Present illness; use the OPQRSTA approach to cover all aspect of information.
- Past illness ; Lists childhood illnesses and lists adult illnesses with dates for at least four categories:

## Cont'd history

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- ***Family history***- any member of the family had a problem or any inherited conditions in the family
- Malignant hyperthermia (MH)
- Hemophilia
- Unknown death
- Delayed awakening
- Sickle cell disease

## Cont'd..... history

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### ➤ *Drug history and allergy*

- Allergies to drugs like sulfonamides, penicillin
- Latex allergy ,iodine allergy
- Allergy to shellfish, egg,.....

## Cont'd..... history

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### *True allergic reaction*

- Skin manifestation(prurities, flushing)
- Face swelling
- Shortness of breath
- Wheezing
- Vascular collapse

# Personal and social history

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## Smoking

- How long the patient has been smoking and their conception of cigarettes per day

The adverse effect of smoking includes;

- Decreased oxygen carriage due to raised carboxyhemoglobin levels
- Increase mucus production and decreased clearance.
- Pulmonary Hypertension; due to the destruction of pulmonary capillaries in the alveolar septa.

## cont'd...Personal and social history

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Smokers have a significant increase in

- Post operative chest infection
- Chronic lung disease
- Carcinoma

Stop smoking for;

- ❖ Eight weeks improves the airways
- ❖ Two weeks reduce irritability
- ❖ Less than 24 hours before anesthesia decrease carboxy hemoglobin level

## Cont'd...Personal and social history

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### Alcohol

- Induction of liver enzymes and tolerance to anesthetic drugs

### Drugs

- Ask for recreational drug types, frequency, and route of administration

### Pregnancy

- ❖ Date of last menstrual

# Previous anesthetic and operations history

## The patient should be asked

- Inherited or family disease
- Whether they have experienced any difficulty previous anesthesia like nausea and vomiting

## *The record of previous anesthesia should be checked to rule out*

- Any difficulties with intubation
- Response to sedative or analgesics
- Anesthetic agents and pre medications
- Halothane exposure
- Post operative complication

Patient's name \_\_\_\_\_ Age \_\_\_\_\_ Sex \_\_\_\_\_ Date of surgery \_\_\_\_\_

Planned operation \_\_\_\_\_ Surgeon \_\_\_\_\_

Primary care doctor/phone # \_\_\_\_\_ Other physicians/phone #s \_\_\_\_\_

1. Please list **all operations** (and approximate dates)

- |          |          |
|----------|----------|
| a. _____ | d. _____ |
| b. _____ | e. _____ |
| c. _____ | f. _____ |

2. Please list any **allergies** to medicines, latex or other (and your reactions to them)

- |          |          |
|----------|----------|
| a. _____ | c. _____ |
| b. _____ | d. _____ |

3. Please list **all medications** you have taken in the last month (include over-the-counter drugs, inhalers, herbals, dietary supplements and aspirin)

- | Name of Drug | Dose and How Often | Name of Drug | Dose and How Often |
|--------------|--------------------|--------------|--------------------|
| a. _____     | _____              | f. _____     | _____              |
| b. _____     | _____              | g. _____     | _____              |
| c. _____     | _____              | h. _____     | _____              |
| d. _____     | _____              | i. _____     | _____              |
| e. _____     | _____              | j. _____     | _____              |

(Please check YES or NO and circle specific problems)

4. Have you taken steroids (prednisone or cortisone) in the last year?
5. Have you ever smoked? (Quantify in \_\_\_\_\_ packs/day for \_\_\_\_\_ years)
- Do you still smoke?
- Do you drink alcohol? (If so, how much?) \_\_\_\_\_
- Do you use or have you ever used any illegal drugs? (we need to know for your safety)
6. Can you walk up one flight of stairs without stopping?
7. Have you had any problems with your heart? (**circle**) (chest pain or pressure, heart attack, abnormal ECG, skipped beats, heart murmur, palpitation, heart failure [fluid in the lungs], require antibiotics before routine dental care)
8. Do you have high blood pressure?
9. Have you had any problems with your lungs or your chest? (**circle**) (shortness of breath, emphysema, bronchitis, asthma, TB, abnormal chest x-ray)
10. Are you ill now or were you recently ill with a cold, fever, chills, flu or productive cough?

YES

NO

☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

(Please check YES or NO and circle specific problems)

11. Have you or anyone in your family had serious bleeding problems? **(circle)** (prolonged bleeding from nosebleed, gums, tooth extractions, or surgery)

YES

NO

☐☐

12. Have you had any problems with your blood (anemia, leukemia, sickle cell disease, blood clots, transfusions)?

☐☐

13. Have you ever had problems with your: **(circle)**

Liver (cirrhosis, hepatitis, jaundice)?

Kidney (stones, failure, dialysis)?

Digestive system (frequent heartburn, hiatus hernia, stomach ulcer)?

Back, neck or jaws (TMJ, rheumatoid arthritis)?

Thyroid gland (underactive or overactive)?

☐☐☐☐☐☐☐☐☐☐☐☐

14. Have you ever had: **(circle)**

Seizures, epilepsy, or fits?

Stroke, facial, leg or arm weakness, difficulty speaking?

Cramping pain in your legs with walking?

Problems with hearing, vision or memory?

☐☐☐☐☐☐☐☐☐☐

15. Have you ever been treated for cancer with chemotherapy or radiation therapy? **(circle)**

☐☐

16. Women: Could you be pregnant?

☐☐

Last menstrual period began: \_\_\_\_\_

17. Have you ever had problems with anesthesia or surgery? **(circle)** (severe nausea or vomiting, malignant hyperthermia (in blood relatives or self), prolonged drowsiness, anxiety, breathing difficulties, or problems during placement of a breathing tube)

☐☐

18. Do you have any chipped or loose teeth, dentures, caps, bridgework, braces, problems opening your mouth, swallowing or choking? **(circle)**

☐☐

19. Do your physical abilities limit your daily activities?

☐☐

20. Do you snore?

☐☐

21. Please list any medical illnesses not noted above:

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

22. Additional comments or questions for nurse or anesthesiologist?

\_\_\_\_\_

\_\_\_\_\_

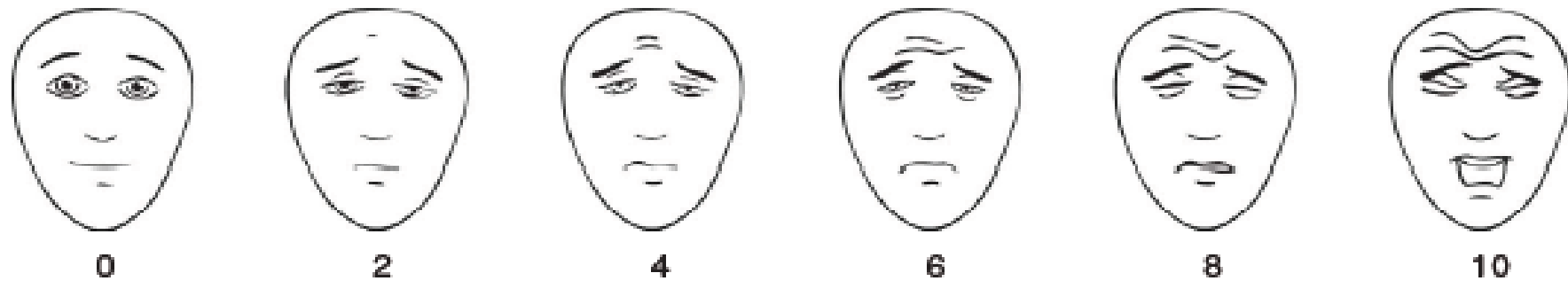
# Physical examination

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Physical examination should build on the information gathered during the history.

- Start from general survey of the patient; Is the patient acutely sick looking or chronically? Acutely sick looking: in pain or respiratory distress, Chronically: prominent zygoma, How the patient walks, body habitus.

# Asses the pain scale of the patient



**FIGURE 4-9.** Pain rating scale. Explain to the person that each face is for a person who feels happy because he has no pain (hurt) or sad because he has some or a lot of pain. Face 0 is very happy because he doesn't hurt at all. Face 2 hurts just a little bit. Face 4 hurts a little more. Face 6 hurts even more. Face 8 hurts a whole lot. Face 10 hurts as much as you can imagine, although you don't have to be crying to feel this bad. Ask the person to choose the face that best describes how he is feeling. (Faces Pain Scale—Revised (FPS-R). [www.iasp-pain.org/fpar](http://www.iasp-pain.org/fpar). Copyright © 2001, International Association for the Study of Pain®. Reproduced with permission.)

**TABLE 5-18** Classification of Nutritional Status by Body Mass Index in Adults

| Body Mass Index (kg/m <sup>2</sup> ) | Nutritional Status      |
|--------------------------------------|-------------------------|
| <16.0                                | Severely malnourished   |
| 16.0-16.9                            | Moderately malnourished |
| 17.0-18.4                            | Mildly malnourished     |
| 18.5-24.9                            | Normal                  |
| 25.0-29.9                            | Overweight              |
| 30.0-34.9                            | Obese (class I)         |
| 35.0-39.9                            | Obese (class II)        |
| ≥40                                  | Obese (class III)       |

# Physical Examination

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Examination Techniques include:

*1. Inspection:*

*2. Palpation:*

*3. Percussion:*

*4. Auscultation:*

- These four techniques are foundation of physical examination.

***Inspection:*** The visual examination of the body using the eyes and a lighted instrument if needed. The sense of smell may also be used.

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***Palpation:*** *the* act of feeling with the hand; the application of the fingers with light pressure to the surface of the body for the purpose of determining the condition of the parts beneath in physical diagnosis

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***Percussion:*** in physical examination, striking a part of the body with short, sharp blows of the fingers in order to determine the size, position, and density of the underlying parts by the sound obtained. Percussion is most commonly used on the chest and back for examination of the heart and lungs

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**Auscultation:** listening for sounds produced within the body, chiefly to assess the condition of the thoracic or abdominal organs and vessels.



# The comprehensive physical exam

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- General Survey
  - weight change, weakness, fatigue, fever
- HEENT
  - Head – headache
  - Ears – ache, deafness, discharge
  - Eyes – pain, itching, photophobia, lacrimation
  - Nose – epistaxis, discharge
  - Mouth & throat – bleeding gums, sore throat
- LN – any swelling , lump in breast

# ....The comprehensive physical exam

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- Respiratory system
  - Cough, sputum, chest pain, wheezing
- Cardiovascular system
  - Dyspnea, palpitation, orthopnea ,leg swelling
- Gastrointestinal system
  - Nausea, vomiting ,heartburn, bowel habits, cramps
- Genitourinary system
  - Flank pain, dysuria, urgency, frequency(D:N)
  - Menstrual problems

# .....The comprehensive physical exam

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- Integumentary system
  - Rashes, itching
- Allergy
  - Drug sensitivity, eczema
- Musculoskeletal system
  - Joint pain, swelling, varicose vein
- Central nervous system
  - Seizures, poor memory, failure to use limbs, fainting

# Vital signs

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- ❖ Pulse rate: The number of pulsations felt in one minute
- ❖ Pulse Rhythm: The pattern and equality of intervals between beats
- ❖ Pulse Quality: The strength, which can be weak, thready, strong, or bounding.
- ❖ Blood pressure: The force of blood against arteries walls as the heart contracts and relaxes
  - ✓ Systolic—force of blood against the arteries when ventricles *contract*
  - ✓ Diastolic—force of blood against arteries when ventricles *relax*

# How to Check Your Pulse



**1**

**Turn one arm, so your palm is facing up.**



**2**

**Place the middle three fingers of the other arm in the upper groove of the upturned wrist, below where your thumb connects to your wrist.**



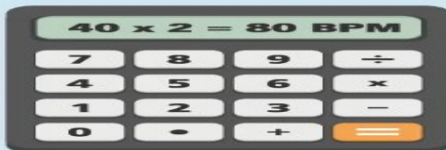
**3**

**Press firmly. You should feel a subtle pulsing beating or throbbing sensation. If you can't feel anything, apply more pressure with the tips of your fingers.**



**4**

**Count each pulse you feel for 30 seconds. (A clock or stopwatch is helpful.)**



**5**

**Double the number of pulses you counted. This equals your heart rate in beats per minute (bpm). If you counted 40 pulses, multiply that by two. Your heart rate is 80 bpm.**

## Normal Heart Rates by Age Range

| Age Range         | Heart Rate (BPM: beats per min) |
|-------------------|---------------------------------|
| Newborn           | 100 – 160                       |
| 0 – 5 months      | 90 – 150                        |
| 6 – 12 months     | 80 – 140                        |
| 1 – 3 years       | 80 – 130                        |
| 3 – 5 years       | 80 – 120                        |
| 6 – 10 years      | 70 – 100                        |
| 11 – 14 years     | 60 – 105                        |
| 15 years or older | 60 – 100                        |

# Blood Pressure Ranges

| BLOOD PRESSURE CATEGORY | SYSTOLIC<br>(mm Hg) | DIASTOLIC<br>(mm Hg) |
|-------------------------|---------------------|----------------------|
| Healthy                 | less than 120       | and less than 80     |
| Elevated                | 120–129             | and less than 80     |
| Stage 1 hypertension    | 130–139             | or 80–89             |
| Stage 2 hypertension    | 140 or higher       | or 90 or higher      |
| Hypertension crisis     | over 180            | or over 120          |

healthline

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❖ Respiration

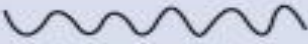
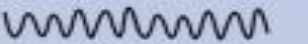
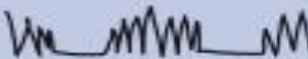
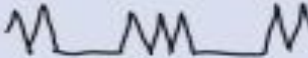
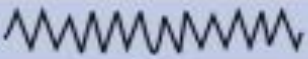
Rate...The number of times a patient breathes in one minute

Effort.... How hard a patient works to breathe

Pattern.... The depth and pattern of breathing.

❖ Body temperature: body works hard to maintain a temperature of approximately 98.6 degrees Fahrenheit (37.0°C).

**Table 1. Breathing patterns**

| Pattern                                                                             | Condition     | Description                                                                                                    |
|-------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------|
|    | Eupnoea       | Normal breathing rate and pattern                                                                              |
|    | Tachypnoea    | Increased respiratory rate                                                                                     |
|    | Bradypnoea    | Decreased respiratory rate                                                                                     |
|    | Apnoea        | Absence of breathing                                                                                           |
|    | Hyperpnoea    | Increased depth and rate of breathing                                                                          |
|    | Cheyne-Stokes | Gradual increases and decreases in respirations with periods of apnoea                                         |
|    | Biot's        | Abnormal breathing pattern with groups/clusters of rapid respiration of equal depth and regular apnoea periods |
|  | Kussmaul's    | Tachypnoea and hyperpnoea                                                                                      |
|  | Apneustic     | Prolonged inspiratory phase with a prolonged expiratory phase                                                  |

## Systemic review

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Special attention to the evaluation of cardiovascular and respiratory systems.

### Air way

The anesthetist needs to recognize the potential for difficulty in maintaining a patent airway with a mask, a laryngeal mask airway, or in the ability to place an endotracheal tube when the patient is under general anesthesia.

The ability to review previous anesthetic records is especially useful in uncovering an unsuspected “difficult airway” or to confirm previous uneventful tracheal intubations, noting whether the patient’s body habitus or airway anatomy has changed in the interim.

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- Mallampati classification
  - Class I full view of soft palate, Uvula ,tonsillar pillars
  - Class II Soft palate and upper portion of Uvula
  - Class III Soft palate
  - Class IV Hard palate

## The Mallampati Score



**CLASS I**  
Complete  
visualization of  
the soft palate



**CLASS II**  
Complete  
visualization  
of the uvula



**CLASS III**  
Visualization  
of only the  
base of the uvula



**CLASS IV**  
Soft palate  
is not  
visible at all

# Cardiovascular system assessment:-

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- Ask symptoms like chest pain, exertional dyspnea, body swelling, orthopnea, PND, syncope
- History of known illnesses like hypertension, myocardial infarction, heart failure, valvular heart disease
- Physical examination:- Check pulse rate on radial artery (normal 60-100 bpm), check for rhythm.

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- Check for pulses in other arterial areas.
  - Check jugular venous distension
  - Check BP on both arms (normal 90-130/60-80)
  - Auscultation for heart sounds, any murmur
  - Auscultate over carotid arteries

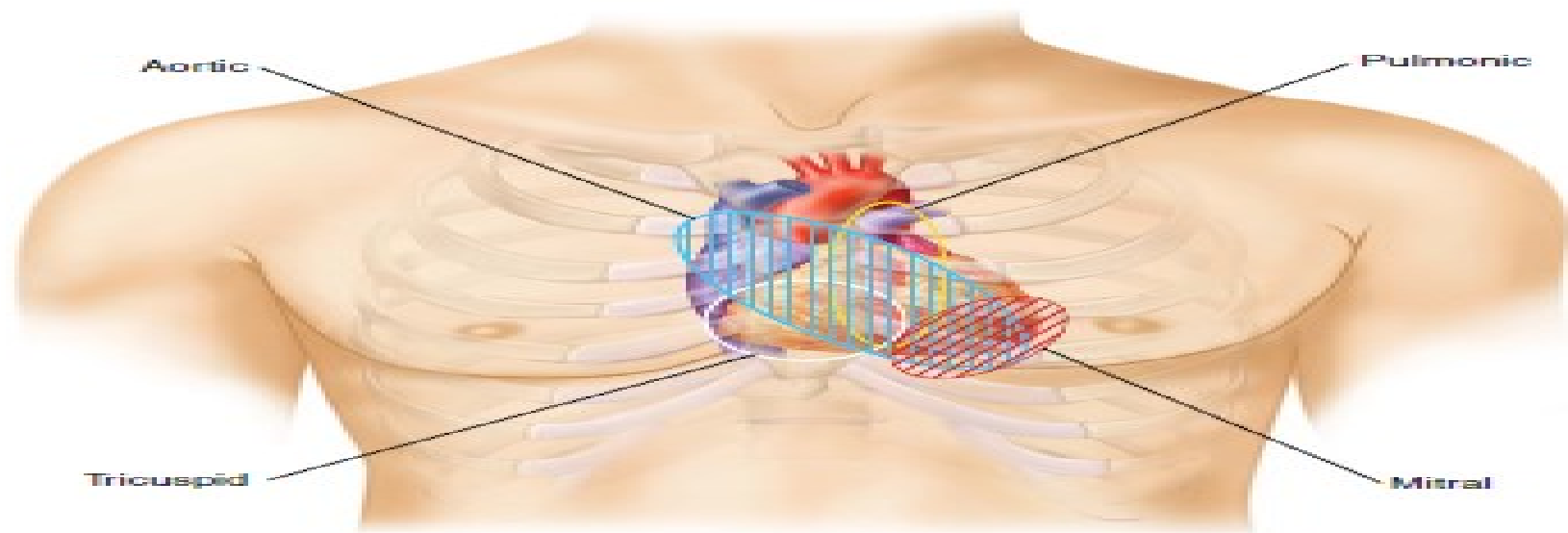
### Chest Wall Location

Right 2nd interspace to the apex  
Left 2nd and 3rd interspaces close to the sternum, but also at higher or lower levels  
At or near the lower left sternal border  
At and around the cardiac apex

### Typical Origin of Sounds and Murmurs

Aortic valve  
Pulmonic valve  
Tricuspid valve  
Mitral valve

*These areas overlap, as illustrated in Figure 9-13. Integrating the auscultatory location with the timing of the sound or murmur, either systole or diastole, is an important first step in identifying sounds and murmurs correctly, and often leads to accurate bedside diagnosis when integrated with other cardiac findings.*



**FIGURE 9-13.** Listening areas on the chest wall.

# Preoperative pulmonary assessment

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- History of tobacco use, shortness of breath, cough, wheezing, stridor, and snoring or sleep apnea.
- The patient should also be questioned regarding the presence or recent history of an upper respiratory tract infection.
- Known chronic illnesses like asthma, COPD, history of TB...

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Markedly impaired pulmonary function is likely in patients who have the following:

- Any chronic disease that involves the lung
- Smoking history, persistent cough, and/or wheezing
- Chest wall and spinal deformities
- Morbid obesity
- Requirement for single-lung anesthesia or lung resection
- Severe neuromuscular disease

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- Wheezing ,Strider, consolidation ,effusion and restrictive lung disease
  - Restrictive disease is characterized by proportional decreases in all lung volumes.
  - The decreased FRC produces low lung compliance and also results in arterial hypoxemia because of low mismatching.
  - Patients with this disease typically breathe rapidly and shallowly.

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Physical examination:-

- Check respiratory rate (normal 14-20)
- Use of accessory muscles
- Shape of the chest: any deformity..
- Look for cyanosis: tongue and extremities

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- Auscultation: check for quality of air entry and added sounds.
  - Decreased air entry: can be due to pleural effusion, mass, consolidation
  - Wheeze: due to asthma mostly
  - Crackles: due to pneumonia, pulmonary edema



**FIGURE 8-19.** Percuss and auscultate in a "ladder" pattern.



**FIGURE 8-14.** Assess lung expansion.



**FIGURE 8-24.** Palpate and percuss in a "ladder" pattern.

# Pre operative assessment of Neurologic System

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- Neurologic system assessment in the apparently healthy patient can be accomplished through simple observation.
- The patient's ability to answer health history questions practically indicates a normal mental status
- Questions can be directed regarding a history of stroke, symptoms of cerebrovascular disease, seizures, pre-existing neuromuscular disease, or nerve injuries.
- The neurologic examination may be cursory in healthy patients or extensive in patients with coexisting disease.
- Testing of strength, reflexes, and sensation may be important in patients for whom the anesthetic plan or surgical procedure may result in a change in condition.

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- Headache
- Dizziness or vertigo
- Weakness (generalized, proximal, or distal)
- Numbness, abnormal or absent sensation
- Fainting and blacking out (near-syncope and syncope)
- Seizures
- Tremors or involuntary movements

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- History of loss of consciousness, trauma, stroke, change of vision....
  - Physical examination:- Mental Status: check orientation to time, place, and person
  - Sensory: pain, temperature, light touch, vibration
  - Pupils: size and position, response to light..



### Small or Pinpoint Pupils

*Bilaterally small pupils* (1–2.5 mm) suggest damage to the sympathetic pathways in the hypothalamus, or metabolic encephalopathy, a diffuse failure of cerebral function that has many causes, including drugs. Light reactions are usually normal.

*Pinpoint pupils* (<1 mm) suggest a hemorrhage in the pons, or the effects of morphine, heroin, or other narcotics. The light reactions may be seen with a magnifying glass.



### Midposition Fixed Pupils

Pupils that are in the *midposition or slightly dilated* (4–6 mm) and are *fixed to light* suggest structural damage in the midbrain.



### Large Pupils

*Bilaterally fixed and dilated pupils* may be due to severe anoxia and its sympathomimetic effects, as seen after cardiac arrest. They may also result from atropine-like agents, phenothiazines, or tricyclic antidepressants.

*Bilaterally large reactive pupils* may be due to cocaine, amphetamine, LSD, or other sympathetic nervous system agonists.



### One Large Pupil

A pupil that is *fixed and dilated* warns of herniation of the temporal lobe, causing compression of the oculomotor nerve and midbrain. A single large pupil is commonly seen in diabetic patients with infarction of CN III.

# Glasgow coma scale(GCS):-

## Activity

## Score

### Eye Opening

None

1 = Even to supraorbital pressure

To pain

2 = Pain from sternum/limb/supraorbital pressure

To speech

3 = Nonspecific response, not necessarily to command

Spontaneous

4 = Eyes open, not necessarily aware

### Motor Response

None

1 = To any pain; limbs remain flaccid

Extension

2 = Shoulder adducted and shoulder and forearm internally rotated

Flexor response

3 = Withdrawal response or assumption of hemiplegic posture

Withdrawal

4 = Arm withdraws to pain, shoulder abducts

Localizes pain

5 = Arm attempts to remove supraorbital/chest pressure

Obeys commands

6 = Follows simple commands

### Verbal Response

None

1 = No verbalization of any type

Incomprehensible

2 = Moans/groans, no speech

Inappropriate

3 = Intelligible, no sustained sentences

Confused

4 = Converses but confused, disoriented

Oriented

5 = Converses and is oriented

**TOTAL (3-15)<sup>a</sup>**

<sup>a</sup>Interpretation: Patients with scores of 3-8 usually are considered to be in a coma.

Motor: bulk, symmetry, tone, power, deep tendon reflex.

### **Scale for Grading Muscle Strength**

Muscle strength is graded on a 0 to 5 scale:

- 0—No muscular contraction detected
- 1—A barely detectable flicker or trace of contraction
- 2—Active movement of the body part with gravity eliminated
- 3—Active movement against gravity
- 4—Active movement against gravity and some resistance
- 5—Active movement against full resistance without evident fatigue. This is normal muscle strength.

Source: Medical Research Council. Aids to the examination of the peripheral nervous system. London: Bailliere Tindall, 1986.

# Gastrointestinal(GI) system assessment

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- History of alcohol use, previous abdominal surgery,...
  - Fluid and electrolyte imbalance
  - Liver cirrhosis
  - Coagulopathies
  - GER
  - Rapid sequence induction and prophylaxis preoperatively

Physical examination:- palpate for mass, tenderness, enlarged organ like liver and spleen.

## **Gastrointestinal Disorders**

Abdominal pain, acute and chronic  
Indigestion, nausea, vomiting including blood (*hematemesis*), loss of appetite (*anorexia*), early satiety

Difficulty swallowing (*dysphagia*) and/or painful swallowing (*odynophagia*)

Change in bowel function

Diarrhea, constipation

Jaundice

# Pre operative assessment of Endocrine

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- Endocrine ;Ask known illnesses like diabetes, adrenal insufficiency, thyroid disorders
- Symptoms: fatigue, palpitation, sweating, weight change, mood swings, skin color change...

# Genitourinary system assessment

## **Urinary and Renal Disorders**

Suprapubic pain

Difficulty urinating (*dysuria*), urgency, or frequency

Hesitancy, decreased stream in males

Excessive urination (*polyuria*) or excess urination at night (*nocturia*)

Urinary incontinence

Blood in the urine (*hematuria*)

Flank pain and ureteral colic

## . Musculoskeletal system assessment

- Joint pain: articular or extra-articular, acute or chronic, inflammatory or noninflammatory, localized or diffuse
- Joint pain: associated constitutional symptoms and systemic manifestations from other organ systems
- Neck pain
- Low back pain

## Assessing the Four Signs of Inflammation

- *Swelling.* Palpable swelling may involve: (1) the synovial membrane, which can feel boggy or doughy; (2) effusion from excess synovial fluid within the joint space; or (3) soft tissue structures, such as bursae, tendons, and tendon sheaths.
- *Warmth.* Use the backs of your fingers to compare the involved joint with its unaffected contralateral joint, or with nearby tissues if both joints are involved.
- *Redness.* Redness of the overlying skin is the least common sign of inflammation near the joints and is usually seen in more superficial joints like fingers, toes, and knees.
- *Pain or tenderness.* Try to identify the specific anatomic structure that is tender.

# Inspection of the Spine

## View of Patient

### From the side

Inspect the cervical, thoracic, and lumbar curves.

Cervical  
concavity

Thoracic  
convexity

Lumbar  
concavity



## Neck Range of Motion

| Neck Movement   | Primary Muscles Affecting Movement                          | Patient Instructions                                 |
|-----------------|-------------------------------------------------------------|------------------------------------------------------|
| Flexion         | Sternocleidomastoid, scalene, prevertebral muscles          | <i>"Bring your chin to your chest."</i>              |
| Extension       | Splenius capitis and cervicis, small intrinsic neck muscles | <i>"Look up at the ceiling."</i>                     |
| Rotation        | Sternocleidomastoid, small intrinsic neck muscles           | <i>"Look over one shoulder, and then the other."</i> |
| Lateral Bending | Scalenes and small intrinsic neck muscles                   | <i>"Bring your ear to your shoulder."</i>            |

# Laboratory and Radiographic examination

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- Complete Blood Count, Hemoglobin, and Hematocrit ;Major surgery
- Chronic cardiovascular, pulmonary, renal, hepatic disease or malignancy
- Known or suspected anemia, bleeding diathesis, or myelosuppression less than one year of age....
- Renal Function Testing(Electrolyte, creatinine and blood urea nitrogen)
- Coagulation Testing (International normalizing ratio(INR), activated partial thromboplastin time(aPTT):- Anti-coagulant therapy, Bleeding diathesis, Liver disease....

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- The Electrocardiogram; Heart disease, hypertension, diabetes
  - Other risk factors for cardiac disease(may include age) Sub-arachinoid or intracranial hemorrhage, cerebrovascular accident.....
  - The Chest Radiograph ;cardiac and pulmonary disease, malignancy.....
  - Fasting glucose:- Diabetes (should be repeated on the day of surgery)

# Respiratory disease

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## **Spirometry:-**

- Diagnosis of a disease process
- Monitoring the response to therapy
- Preoperative assessment for lung resection, cardiac surgery or non-cardiothoracic surgery
- Evaluating disease prognosis.

# Pregnancy test

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- Pregnancy Test ;On the day of surgery, ask all women of childbearing potential whether there is any possibility they could be pregnant.
- Make sure women who could possibly be pregnant are aware of the risks of the anesthetic and the procedure to the fetus.

**Table 34-7 -- Diagnosis-Based Preoperative Testing**

| Preoperative Diagnosis      | ECG                                                     | CXR | Hct/Hb | CBC | Lytes | Renal | Glucose | Coag | LFTs | Drug Levels | Ca |
|-----------------------------|---------------------------------------------------------|-----|--------|-----|-------|-------|---------|------|------|-------------|----|
| Cardiac disease             |                                                         |     |        |     |       |       |         |      |      |             |    |
| MI history                  | X                                                       |     |        |     | ±     |       |         |      |      |             |    |
| Stable angina               | X                                                       |     |        |     | ±     |       |         |      |      |             |    |
| CHF                         | X                                                       | ±   |        |     |       |       |         |      |      |             |    |
| HTN                         | X                                                       | ±   |        |     | X *   | X     |         |      |      |             |    |
| Chronic atrial fibrillation | X                                                       |     |        |     |       |       |         |      |      | X [†]       |    |
| PVD                         | X                                                       |     |        |     |       |       |         |      |      |             |    |
| Valvular heart disease      | X                                                       | ±   |        |     |       |       |         |      |      |             |    |
| Pulmonary disease           |                                                         |     |        |     |       |       |         |      |      |             |    |
| Emphysema                   | X                                                       | ±   |        |     |       |       |         |      |      | X [‡]       |    |
| Asthma                      | (PFTs only if symptomatic; otherwise no tests required) |     |        |     |       |       |         |      |      |             |    |
| Chronic bronchitis          | X                                                       | ±   |        | X   |       |       |         |      |      |             |    |
| Diabetes                    | X                                                       |     |        |     | ±     | X     | X       |      |      |             |    |
| Hepatic disease             |                                                         |     |        |     |       |       |         |      |      |             |    |
| Infectious hepatitis        |                                                         |     |        |     |       |       |         | X    | X    |             |    |
| Alcohol/drug induced        |                                                         |     |        |     |       |       |         | X    | X    |             |    |
| Tumor infiltration          |                                                         |     |        |     |       |       |         | X    | X    |             |    |
| Renal disease               |                                                         |     | X      |     | X     | X     |         |      |      |             |    |
| Hematologic disorders       |                                                         |     |        | X   |       |       |         |      |      |             |    |
| Coagulopathies              |                                                         |     |        | X   |       |       |         | X    |      |             |    |
| CNS disorders               |                                                         |     |        |     |       |       |         |      |      |             |    |
| Stroke                      | X                                                       |     |        | X   | X     |       | X       |      |      | X           |    |
| Seizures                    | X                                                       |     |        | X   | X     |       | X       |      |      | X           |    |
| Tumor                       | X                                                       |     |        | X   |       |       |         |      |      |             |    |

# Preoperative risk assessment

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- The current ASA risk classification system was developed in 1941 by Meyer [Saklad](#) at the request of the ASA.
- ASA physical status (ASA-PS) 2 patients are at higher risk than ASA-PS 1 patients, but only if undergoing the same operation.

# Preoperative Risk Assessment

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## ***ASA Class 1***

No organic, physiologic, biochemical, or psychiatric disturbance

e.g. healthy patient, no medical problem

## ***ASA Class 2***

Mild-to-moderate systemic disturbance that may not be related to the reason for surgery

e.g. controlled hypertension

## ***ASA Class 3***

Severe systemic disturbance that may or may not be related to the reason for surgery

e.g. Emphysema.

# ...Preoperative Risk Assessment

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## ***ASA Class 4***

Severe systemic disturbance that is life-threatening with (without )surgery  
e.g. Unstable angina.

## ***ASA Class 5***

Moribund patient who has little chance of survival but is submitted to surgery as a last resort  
(resuscitative effort)

## ***ASA class 6***

A declared brain-dead patient whose organs are being removed for donor Purposes.

## ***Emergency operation (E)***

Any patient in whom an emergency operation is required

# Preoperative preparation and premedication

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## *Procedure-related reasons:*

1. Antibiotic prophylaxis to prevent infections.
2. Gastric prophylaxis (to minimize the risk of gastric aspiration during anesthesia).
3. Corticosteroid coverage in patients who are immunosuppressed.
4. To avoid undesired reflexes arising during a procedure (e.g., vagal reflex during eye surgery).
5. Anticholinergic agents to decrease oral secretions and facilitate a planned awake intubation with a fiberoptic bronchoscope.

## ...preoperative preparation and pre medication

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### *Coexisting Diseases:*

1. To continue the patient's own medications for coexisting diseases. (e.g., beta blockers, antihypertensive medications, etc.)

As a general rule, all cardiac and pulmonary medications and most other necessary medications should be taken with sips of water at the usual time, up to and including the day of surgery.

Possible exceptions to this include Coumadin, ASA and NSAID's, insulin (adjustment of the dose is needed on the day of surgery), oral hypoglycemic and antidepressants.

2. To optimize the patients status prior to the procedure. (e.g., bronchodilators, beta blockers, antibiotics.

# Pre-operative orders

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## **NPO Status:-**

- Pulmonary aspiration of gastric contents, even 30-40ml, is associated with significant morbidity and mortality.
- Factors predisposing to regurgitation and pulmonary aspiration include inadequate anesthesia, pregnancy, obesity, difficult airway, emergency surgery, full stomach, and altered gastrointestinal motility
- Fasting before anesthesia aims to reduce the volume of gastric contents and hence the risk should aspiration occur.

# Fasting Recommendations

| • Ingested Material | Minimum fasting period applied to all ages (hr) |
|---------------------|-------------------------------------------------|
| Clear liquids.....  | 2                                               |
| Breast milk.....    | 4                                               |
| Infant formula..... | 6                                               |
| Nonhuman milk.....  | 6                                               |
| Light meal.....     | 6                                               |
| Heavy meal.....     | 8                                               |

# Medications to continue and discontinue:-

---

## Drugs to continue:-

1. anti-hypertensive medications
2. Thyroid medications
3. Cardiac medications
4. Psychiatric medications
5. Anti-convulsant medications
6. Asthma medications
7. Steroid
8. Aspirin

## Drugs to discontinue:-

1. Insulin
2. Oral hypoglycemic agents
3. Diuretics
4. NSAID's
5. warfarin

# Pre operative medication

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- Preoperative medication consists of psychological preparation and pharmacological preparation.
- Goals of premedication:-
  - Relief of anxiety
  - Sedation
  - Amnesia
  - Analgesia

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- Drying of airway secretion
  - Prevention of autonomic reflex
  - Reduction of gastric fluid volume and increased pH
  - Antiemetic effect
  - Reduction of the anesthetic requirement
  - Facilitation of smooth induction of anesthesia
  - Prophylaxis against allergic reactions

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- Psychological preparation of the patient involves a pre-operative visit and interview with the patient and family members.
  - The anesthetist should explain anticipated events and the proposed anesthetic management in an effort to reduce anxiety.
  - An informative and comforting pre-operative visit may replace many milligram of sedative medications.

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- **Pharmacological preparation:-**
  - Drugs selected for preoperative medication are typically administered orally or IM in patients room 1-2 hours before the anticipated administration of anesthesia.
  - For out patient surgery preoperative drugs can be administered IV immediately in the operation room.

# Determination of choice of drug:-

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- Patient's age and weight
- Cardiac status
- Level of anxiety
- Tolerance of depressant drugs
- Drug allergies
- Type of surgery (emergency or elective)

### Commonly used premedication

| <u>Medication</u> | <u>Administration route</u> | <u>Dose(mg)</u>   |
|-------------------|-----------------------------|-------------------|
| Lorazepam         | oral,IV                     | 0.5-4             |
| Midazolam         | IV                          | titration1.0-2.5  |
| Fentanyl          | IV                          | titration25-100µg |
| Morphine          | IV                          | titration1-2.5    |
| Meperidine        | IV                          | titration10-25    |
| Cimetidin         | oral,IV                     | 150-300           |
| Ranitidine        | oral                        | 50-200            |
| Methochlorpromide | oral ,IV                    | oral 10, IV5-10   |
| Glycopyrrolate    | IV                          | 0.1-0.2           |
| Atropine          | IV                          | 0.3-0.4           |
| Scopolamine       | IV                          | 0.1-0.4           |

## Plan of anesthetic technique

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1. Is the patient's condition optimal?
2. Are there any problems which require consultation or special tests?
3. Is there an alternative procedure which may be more appropriate?
4. What are the plans for post- operative management of the patient?
5. What premedication if any is appropriate?

# Documentation on anesthesia record

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Recording information on anesthesia is important.

- The anesthesia provider is responsible for the patient from the time they enter the operating room through the recovery period.
- Maintaining a continuous record will help the anesthesia provider remain vigilant during the operation time
- Documentation should be neat and legible.
- A well documented anesthesia record is useful for future anesthetics ,any complication, and guiding care of patient.
- Review diagnostic studies including ECG ,CXR ,and document the results.
- The next step involves the patient interview and a physical exams.
- It consist of a systemic inquiry concerning the patient health history ,ASA classification ,air way assessment finding and anesthetic plan

# Pre anesthesia evaluation document

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- The first item that should be documented is identification of the patient.
- Date of the surgery, name ,proposed surgical procedure name of the surgeon family contact, name of the anesthesia provider ward and bed number ,height weight and pre operative vital signs should be documented
- Review the patients laboratory values and document the results.

## Anesthesia care documentation

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- If the patient take pre medication document the medication dose and route
- Apply monitors to the patient and record initial set of vital sign.
- Document the rhythm ,heart rate blood pressure and pulse oximetry reading
- Document the intravenous fluid type and the amount infused
- Document the medications that you will administer during the anesthesia time.

## ...Anesthesia care documentation

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- Document ;the patient has been pre oxygenated
- Type of induction
- Use of LMA, mask ,or ETT
- Type of intubation(oral or nasal),ETT size, laryngoscope blade ,size ,any difficulties encountered during intubation presence
- Document that the patient eyes are protected
- If the patient receives regional register type of block, patient position, needle type and size ,dose, volume, and level of block.
- At the end the total amount of drug ,IV fluids ,blood or blood product s urine out put initial vital sign should be documented.

## Cont'd...Anesthesia care documentation chart

| <b>ANESTHESIA RECORD</b>                                                                                                                    |  |  |  |  |  |  |  |  |               |
|---------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|---------------|
| AGE: _____ SEX: M F PRE MED: S U _____ ASA _____                                                                                            |  |  |  |  |  |  |  |  |               |
| DENTITION _____ NPO _____ PROPOSED SURGERY _____                                                                                            |  |  |  |  |  |  |  |  |               |
| <input type="checkbox"/> PT. IDED <input type="checkbox"/> CONSENT ✓ ED <input type="checkbox"/> CHART REVIEWED               SURGEON _____ |  |  |  |  |  |  |  |  |               |
| OPERATION PERFORMED _____                                                                                                                   |  |  |  |  |  |  |  |  |               |
| <b>TIME</b>                                                                                                                                 |  |  |  |  |  |  |  |  | <b>TOTALS</b> |
| OXYGEN                                                                                                                                      |  |  |  |  |  |  |  |  |               |
| NITROUS OXIDE                                                                                                                               |  |  |  |  |  |  |  |  |               |
| HALO/ENFL/ISOFL/DES/SEVO                                                                                                                    |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
| TEMP                                                                                                                                        |  |  |  |  |  |  |  |  |               |
| URINE                                                                                                                                       |  |  |  |  |  |  |  |  |               |
| FLUIDS/BLOOD                                                                                                                                |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |
|                                                                                                                                             |  |  |  |  |  |  |  |  |               |

## ...Anesthesia care documentation chart

| IV # _____                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               | ECG               | <input type="checkbox"/> MONITORED ANESTHESIA CARE<br><input type="checkbox"/> REGIONAL<br><input type="checkbox"/> GENERAL<br>BLADE _____<br>ETT# _____<br>BBS _____<br>CUFF _____ cc.<br><input type="checkbox"/> ATRALUMATIC<br><input type="checkbox"/> CO <sub>2</sub><br><b>COMMENTS:</b><br><br>EYE PROTECTION: _____<br><br>ANESTH. START _____<br>ANESTH. INDUC _____<br>SURG. START _____<br>SURG. END _____<br>ANESTH. END _____<br>ANESTH. NET _____ |  |  |  |  |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|
| BP                                                                                                                                                                                                                                                                                                                                                                                      | V<br>SYS<br>DIA                                                                                               | FIO <sub>2</sub>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| PULSE *                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                               | EtCO <sub>2</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| RESP.                                                                                                                                                                                                                                                                                                                                                                                   | <input checked="" type="radio"/> ASSISTED<br><input type="radio"/> SPONT.<br><input type="radio"/> CONTROLLED | SaO <sub>2</sub>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| <b>MONITORS:</b>                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                               | EtA               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| <input type="checkbox"/> MACHINE <input checked="" type="checkbox"/> ED<br><input type="checkbox"/> OXIMETER<br><input type="checkbox"/> BP. SITE<br><input type="checkbox"/> EKG<br><input type="checkbox"/> FLUID WARMER<br><input type="checkbox"/> PRECORDIAL<br><input type="checkbox"/> TEMP<br><input type="checkbox"/> HUMIDIFIER<br><input type="checkbox"/> BLANKET WARM/COOL |                                                                                                               | Temp              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| <input type="checkbox"/> RAPID TRANSFER<br><input type="checkbox"/> FORCED AIR WARMER<br><input type="checkbox"/> FIO <sub>2</sub><br><input type="checkbox"/> EtCO <sub>2</sub><br><input type="checkbox"/> ESOPH<br><input type="checkbox"/> NERVE STIM<br><input type="checkbox"/> CVP<br><input type="checkbox"/> A-LINE<br><input type="checkbox"/> PA CATHETER                    |                                                                                                               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| VENT.                                                                                                                                                                                                                                                                                                                                                                                   | VT/RR                                                                                                         |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| BLOOD LOSS POSITION                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                               | P                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |
| REMARKS: _____                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |  |  |  |

## Cont'd...Anesthesia care documentation chart

|                         |            |          |            |            |                           |
|-------------------------|------------|----------|------------|------------|---------------------------|
| RECOVERY ROOM           | B.P. _____ | P. _____ | R.R. _____ | TIME _____ | O <sub>2</sub> SAT. _____ |
| CONDITION _____         |            |          |            |            |                           |
| SIGNED (RESIDENT) _____ |            |          |            | M.D.       | PATIENT<br>NAME<br><br>#  |
| SIGNED (STAFF) _____    |            |          |            | M.D.       |                           |
| DATE _____              |            |          |            |            |                           |
| PAGE _____ OF _____     |            |          |            |            |                           |

# Informed consent

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- It is a legal and ethical principle that valid consent must be obtained before starting treatment, physical investigation, or providing personal care for a patient.
- Valid consent implies it is given voluntarily by a competent and informed person not under duress.

# Informed consent

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- To have capacity for consent, the patient must be able to comprehend and remember the information provided, weigh up the risks and benefits of the proposed procedure and consider the consequences of not having the procedure in order to make a balanced decision.
- Consent may be expressed, either written or verbal, or implied.

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- competent young adults over the age of 16yr can give consent for any treatment without obtaining separate consent from a parent or guardian.
  - In an emergency verbal consent by telephone is adequate and essential treatment can be started in the absence of parental authorization if necessary.

## INDIVIDUAL FOLDER

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ቀን: 23/06/22

ስም: B/ Hanisa Asrat

የክርክር ቁጥር: 465069

ዕድሜ: 3

ፆታ: M

አድራሻ: Oremiya

መንታ ክፍል: 312/1

እኔ ስም ከላይ የተጠቀሰው ታካሚ ሊሰራልኝ ለታተመ ለተደራሰበኝ ወይም ለሌላ የህክምና  
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ያሉትም የአንስተኛዎች (ማደንዘዣ ስም) አማራጮች ጥቅማቸውና ጉዳታቸው ሊያጋጥሙ  
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ታካሚ ወይም ታካሚውን የሚወክል ሰው ፊርማ: WA ቀን: 23/06/22

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ቅጽን ያስፈረመው ላለሙያ ስምና ፊርማ: Abile Jeda [Signature] ቀን: 23/06/22

# Assignment

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- 1.Pre operative assessment of Asthma and COPD patients
- 2.Pre operative assessment of Valvular heart disease and Hypertension
- 3.Pre operative assessment of endocrine disease (Diabetic and Thyroid )disease
- 4.Pre operative assessment of liver disease (liver cirrhosis and Hepatitis )
- 5.Pre operative assessment of Renal disease.